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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/675,340	02/18/2022	Yu Fang	SONA-0006-U01-C03	9356

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EXAMINER	
ROY, SANJOY K	

ART UNIT	PAPER NUMBER
2443	

NOTIFICATION DATE	DELIVERY MODE
11/17/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/675,340	Applicant(s) Fang et al.	
	Examiner SANJOY K ROY	Art Unit 2443	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/10/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,3-13,15-19 and 21 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,3-13,15-19 and 21 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 10/10/2023, 10/10/2023, 10/10/2023.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Applicant's arguments and amendments, filed on 10/10/2023 has been entered and carefully considered. Claims 1, 4, 5, 13, 15, and 19 are amended. Claims 2, 14, and 20 are canceled. Claims 1, 3-13, 15-19, and 21 have been examined and rejected.

Response to Amendment and Arguments

3. Applicant's amendments and arguments filed on 10/10/2023 with respect to rejections of claims 1, 3-13, 15-19, and 21 have been considered but are moot in view of the new ground of rejection necessitated by applicant's amendment.

Claim Rejections - 35 USC § 103

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claims 1, 3-13, 15-19, and 21 are rejected under AIA 35 U.S.C. 103 as being unpatentable over Kim et al. (U.S. PGPub 2018/0072250) in view of NEFF et al. (U.S. PGPub 2015/0071115).

As per claims 1, 13 and 19,

Chae teaches a system comprising: a vehicle having a first network and a second network of a different type than the first network (Kim see para 0054 FIG. 1, a communication node constituting a vehicle network may be a gateway, a switch or bridge, gateway 100 is connected with switches 110, 110-1, 110-2, 120, and 130, and configured to connect different networks, the gateway 100 may support connections between a switch which supports a controller area network CAN or, FlexRay, media oriented system transport MOST or local interconnect network LIN and a switch which supports an Ethernet protocol),

and a converged network device (CND) interposed between the first network and the second network and comprising a plurality of ports on the first network; wherein: the CND facilitates communications between the first and the second networks (Kim see para 0072-0076 as shown in fig 5. gateway 500 representing a CND include a controller 510 and a PHY layer 520 and connected to a CAN-based vehicle network 20 using plurality of CAN communication channels 530 that are connected to the controller 510, gateway 500 may include a plurality of communication ports P1, P2, P3, P4, and P5, where first port P1 is connected to an external diagnostic apparatus 10, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30, gateway 500 may relay the communications between the external diagnostic apparatus 10 and a communication node belonging to the vehicle network, and communicate between the first CAN network 20 and second network as the ethernet network 30);

a first port of the plurality mirrors a second port of the plurality (Kim, see para 0106, at step S950, controller 510 may configure the PHY layer so that the PHY layer

performs port mirroring and cause a message received from the external diagnostic apparatus 10 to be mirrored to the target communication node 600, controller 510 receive a diagnostic request message and configure the PHY layer 520 such that the message received from the first port P1 is mirrored to the second port P2 by configuring a destination of the message received from the first port P1 to the second port P2);

Kim fails to exclusively teach and the CND is structured to, based at least in part on a port selection command value from a sensor device external to the vehicle, assign: which port, of the plurality of ports, is the first port; and/or which port, of the plurality of ports, is the second port.

In a similar field of endeavor NEFF teaches and the CND is structured to, based at least in part on a port selection command value (NEFF, see para 0115, method 400 comprises the configuration of an intermediate component 220 of the Ethernet network 110, 200, which lies on the transmission path, to prepare a copy of the data as logging data at step 401and transmitting it to the logging component 230 at step 402, the method 400 comprises the recording 403 of the logging data at the logging component 230 the configuration of an Ethernet switch for a determined logging scenario that the configured Ethernet switch produces a copy of specific data and/or relays specific data to a specific port of the switch, the copy of the data is prepared on the basis of a port mirroring function of the Ethernet switch and configuring the Ethernet switch to transmit the logging data via a determined port to the logging component and configuring the Ethernet switch to reserve a predetermined transmission bandwidth of the determined port for the transmission of the logging data)

from a sensor device external to the vehicle (NEFF, see para 0085 the configuration of the Ethernet switches, the generation and the transmission of the configuration commands is performed via external devices, via a laptop, a tester, or a data logger 26),

assign: which port, of the plurality of ports, is the first port; and/or which port, of the plurality of ports, is the second port (NEFF, see para 0094, 0094 configuration of such actions is performed, explicitly by specifying register values of the Ethernet switch, the configuration can be performed by configuration on the level of port mirroring switch features, the switch register value representing the which port of the switch port is the first port and which port is the destination port o the basis of the switch configuration, it can be established in each switch which data packets are to be relayed to one of the data loggers 260, the selection of the data packets 300 can be performed in this case in an Ethernet switch on the basis of selection of one or more complete Ethernet links 241 which is represented by the source and destination port).

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Kim with the teaching of NEFF, as doing so would provide an efficient method for recording of control unit data or in the transmission of stimulation data to a control unit in automotive Ethernet networks (NEFF see para 0006).

As per claims 3 and 15,

Kim in view of NEFF teaches the system of claim 1, wherein the first port of the plurality mirrors a third port of the plurality (Kim see para 0109 when there are a plurality

of target communication nodes, a message received through the first port P1 may be mirrored so as to be transmitted through a plurality of ports p2 and p3).

As per claim 4,

KIM in view of NEFF teaches the system of claim 1 further comprising: the sensor device external to the vehicle, wherein the sensor device is structured to monitor one or more devices on the first and/or the second networks via communicatively coupling with the first port (Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500).

As per claim 5,

KIM in view of NEFF teaches the system of claim 1 further comprising: the sensor device external to the vehicle, wherein the sensor device is structured to monitor a flow on the first and/or the second networks via communicatively coupling with the first port (Kim, see para 0092, 0093, step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520. The PHY layer 520 may activate the second port P2 connected to the target communication node 600 under control of the controller 510, step S832, the target communication node 600 may receive the security authentication information from the gateway 500, controller 610 of the target communication node 600 may transmit a control command to the PHY layer 620 to activate the sixth port P6 connected to the gateway 500 under control of the controller 610, communications between the external diagnostic apparatus 10 and the target communication node 600 may be allowed by activating the second port P2 and the sixth port P6).

As per claim 6,

KIM in view of NEFF teaches the system of claim 1, wherein the CND is further structured to: assign a first device, of a plurality of devices on the second network, to a first assigned port of the plurality of ports; identify communications data on the second network corresponding to the first device; and transmit the identified communications data corresponding to the first device on the first network via the first assigned port (Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a

diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500);

As per claims 7, 15 and 21,

KIM in view of NEFF teaches system of claim 6, wherein the CND is further structured to: assign a second device, of the plurality of devices on the second network, to a second assigned port of the plurality of ports; identify communications data on the second network corresponding to the second device; and transmit the identified communications data corresponding to the second device on the first network via the second assigned port (Kim see para 0121, In a step S1050, the external diagnostic

apparatus 10 may transmit a diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500);

As per claim 8,

KIM in view of NEFF teaches the system of claim 6, wherein the CND is further structured to: assign a second device, of the plurality of devices on the second network, to the first assigned port; identify communications data on the second network corresponding to the second device; and transmit the identified communications data corresponding to the second device on the first network via the first assigned port(Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a

diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500);

As per claim 9,

KIM in view of NEFF teaches the system of claim 1, wherein the CND further comprises an Ethernet switch that includes the plurality of ports(Kim see para 0054,0075, gateway 100 is connected with switches 110, 110-1, 110-2, 120, and 130, and supports a switch which supports an Ethernet protocol, and control at least one of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 connected to each of the switches where of the end nodes 111, 112, 113, 121, 122, 123, 131, 132,

and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle, gateway 500 include a plurality of communication ports P1, P2, P3, P4, and P5, first port P1 may be connected to an external diagnostic apparatus 1, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30);

As per claims 10 and 16,

KIM in view of NEFF teaches the system of claim 1, wherein the first network is Ethernet based (Kim see para 0072-0076 as shown in fig 5. gateway 500 representing a CND include a controller 510 and a PHY layer 520 and connected to a CAN-based vehicle network 20 using plurality of CAN communication channels 530 that are connected to the controller 510, gateway 500 may include a plurality of communication ports P1, P2, P3, P4, and P5, where first port P1 is connected to an external diagnostic apparatus 10, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30, gateway 500 may relay the communications between the external diagnostic apparatus 10 and a communication node belonging to the vehicle network, and communicate between the first CAN network 20 and second network as the ethernet network 30);

As per claims 11 and 18,

KIM in view of NEFF teaches the system of claim 1, wherein the second network is controller area network (CAN) based (Kim see para 0072-0076 as shown in fig 5.

gateway 500 representing a CND include a controller 510 and a PHY layer 520 and connected to a CAN-based vehicle network 20 using plurality of CAN communication channels 530 that are connected to the controller 510, gateway 500 may include a plurality of communication ports P1, P2, P3, P4, and P5, where first port P1 is connected to an external diagnostic apparatus 10, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30, gateway 500 may relay the communications between the external diagnostic apparatus 10 and a communication node belonging to the vehicle network, and communicate between the first CAN network 20 and second network as the ethernet network 30).

As per claims 12 and 18,

KIM in view of NEFF teaches the system of claim 1, wherein the first network is Ethernet based and the second network is controller area network (CAN) based (Kim see para 0072-0076 as shown in fig 5. gateway 500 representing a CND include a controller 510 and a PHY layer 520 and connected to a CAN-based vehicle network 20 using plurality of CAN communication channels 530 that are connected to the controller 510, gateway 500 may include a plurality of communication ports P1, P2, P3, P4, and P5, where first port P1 is connected to an external diagnostic apparatus 10, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30, gateway 500 may relay the communications between the external diagnostic apparatus 10 and a communication

node belonging to the vehicle network, and communicate between the first CAN network 20 and second network as the ethernet network 30).

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, THIS ACTION IS MADE FINAL. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to SANJOY K ROY whose telephone number is (571)270-0675. The examiner can normally be reached on Mon-Fri 8:30am-5:00pm, EST. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Nicholas R. Taylor can be reached on 571-272-3889. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300. Information regarding the status of an application may be obtained from

the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272- 1000.

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Examiner, Art Unit 2443

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